CS410 Midterm 1 Part (a)

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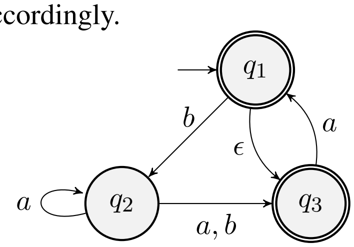
I. Multiple Choice Questions (50 points)

Given the following NFA M_1. Answer questions 1-5 accordingly.



1. What is (are) the start state (s)?

a) q_1
b) q_2
c) q_1, q_3
d) q_1, q_2, q_3
e) None

2. Which of the following can be the input alphabet Σ ?

a) {a}
b) {b}
c) {a, b}
d) {a, ϵ}
e) None

3. What is F?

a) $\{q_1\}$
b) q_1
c) {a, b}
d) {a, b, ϵ}
e) $\{q_1, q_3\}$

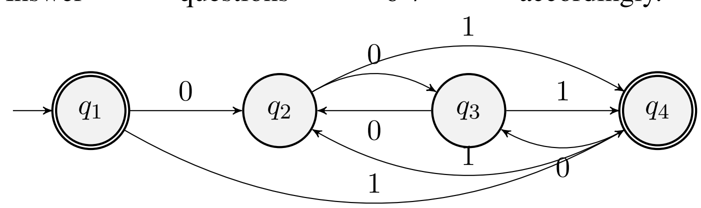
4. How many rows and column does the start transition table have?

a) Rows: 2, Columns: 2
b) Rows: 2, Columns: 3
c) Rows: 3, Columns: 2
d) Rows: 3, Columns: 3
e) None

5. Which of the following is a string that is not accepted by M_1?

a) ϵ
b) a
c) ba
d) bbab
e) bba

Given the following DFA M_2. Answer the questions 6-7 accordingly.



6. Which of the following strings is not accepted by M_2?

a) ϵ
b) 001
c) 011
d) 00101
e) 0010001

7. Which states will be traversed for the input 0000101?

a) $q_1q_2q_3q_4q_3q_4q_3q_4$
b) $q_1q_2q_4q_2q_4q_3q_2q_3$
c) $q_1q_2q_4q_2q_3q_4q_3q_4$
d) $q_1q_2q_3q_2q_3q_4q_3q_4$
e) $q_1q_2q_3q_2q_4q_3q_2q_4$

8. How many states do we need at minimum if we want to design a NFA for the following language $L = \{w|w$ contains 1 in the fourth position from the end$\}$?

a) 4
b) 5
c) 6
d) 7
e) 8

9. If an NFA with 5 states has been transformed into a DFA, at most how many states will it have?

a) 8
b) 16
c) 32
d) 64
e) 128

10. How many states do we need at minimum if we want to design a DFA for the following language $L = \{w|w$ contains exactly two a's$\}$?

a) 1
b) 2
c) 3
d) 4
e) 5

11. If A and B are regular languages with n and m words ($n \leq m$), what is the maximum number of words for the language $L_1 = A \cup B$, $L_2 = A \cap B$?

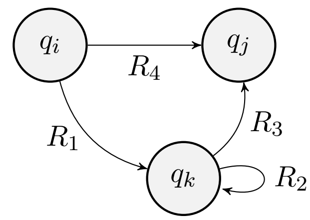
a) m, n
b) n, m
c) m + n, m + n
d) m + n, m
e) m + n, n

12. If A and B are regular, which of the following language is not regular?

a) $A \cup B$
b) $A \cap B$
c) $A \backslash B$
d) $\overline{A}$
e) All of them are regular

13. Given the languages A = $\{$00, 11$\}$ and B = $\{$10, 11$\}$, which of the following is incorrect?

a) $A \cup B = \{00, 11, 10\}$
b) $A \cap B = \{11\}$
c) $A^* = \{00, 11, 0011, 1100, 0000, 1111, \ldots\}$
d) $A \circ A = \{0000, 0011, 1100, 1111\}$
e) $B \circ B = \{1010, 1011, 1110, 1111\}$



14. What will be the corresponding regular expression between node q_i and node q_j, after the node q_k is removed from the given NFA above?

a) $R_1 R_4 R_2 R_3$
b) $R_1 R_2 R_3 \cup R_4$
c) $R_1 R_2^+ R_3 \cup R_4$
d) $R_1 R_2^* R_3 \cup R_4$
e) None of the above

15. Which of the following language is defined with the regular expression $\Sigma^* 01 \Sigma^*$?

a) $A = \{w|w$ contains a 1 after each 0$\}$
b) $B = \{w|w$ contains at least one 0 and one 1$\}$
c) $C = \{w|w$ contains the string 01 as a substring$\}$
d) $D = \{w|w$ starts and ends with 01$\}$
e) $E = \{w|w$ starts with 01$\}$

16. Which of the following language is defined with the regular expression $\Sigma\Sigma\Sigma$?

a) $A = \{w|w$ is a string of odd length$\}$
b) $B = \{w|w$ is a string of even length$\}$
c) $C = \{w|$ length of w is a multiple of 3$\}$
d) $D = \{w|$ length of w is 3$\}$
e) None

17. What is the result of the regular expressions $\emptyset^*$, RR^*?

a) $\emptyset$, R^+
b) $\{\epsilon\}$, R^+
c) $\{\epsilon\}$, R^*
d) $\{\emptyset, \epsilon\}$, R^*
e) $\{\emptyset\}$, R^+

18. Which of the following is the regular expression for the language $A = \{w|w$ is a string of odd length$\}$?

a) Σ^*
b) $\Sigma^*\Sigma^*$
c) $\Sigma^*\Sigma^*\Sigma^*$
d) $\Sigma(\Sigma\Sigma)^*$
e) $(\Sigma\Sigma\Sigma)^*$

19. Which of the following is the regular expression for the language $A = \{w|w$ has at least two 1's$\}$?

a) $\Sigma^{+}1\Sigma^{+}1$
b) $\Sigma^{*}1\Sigma^{*}$
c) $\Sigma^{*}11\Sigma^{*}$
d) $\Sigma^{*}1\Sigma^{*}1$
e) $\Sigma^{*}1\Sigma^{*}1\Sigma^{*}$

20. Which of the following is not a **standard** regular expression operator?

a) $*$
b) $\cap$
c) $\cup$
d) $\circ$
e) All of the above are **standard** regular operators

21. Given the language $A = \{0^n 1^n | n \geq 0\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Let say the selected string is $0^p 1^p$. Which of the following could be the selected string y, that can be pumped?

a) 0^{p+1}
b) 1^p
c) 0011
d) 0000
e) 01

22. Given the language $A = \{0^i 1^j | i > j\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Which of the following strings can be selected string?

a) $0^p 1^{2p}$
b) $0^{p+1} 1^{p+1}$
c) $0^{p+1} 1^p$
d) $0^{p/4} 1^{p/4}$
e) 01

23. Which of the following languages is not regular?

a) $A = \{a^{2n} b^m c | n, m \geq 0\}$
b) $B = \{a^{3n+2} b^{m+1} | m, n \geq 0\}$
c) $C = \{a^n b^m |$ Both n and m are odd$\}$
d) $D = \{a^k b^m c^n | k \geq 2, m \geq 0, n \leq 2\}$
e) All of them are regular

24. Which of the following languages is regular?

a) $A = \{0^n 1^n | n \geq 0\}$
b) $B = \{w|w$ has an equal number of 0s and 1s$\}$
c) $C = \{w|w$ has an equal number of occurrences of 01 and 10 as substrings$\}$
d) $D = \{ww | w \in \{0,1\}^*\}$
e) None

25. Which of the following languages is regular?

a) $A = \{1^n | n \geq 0\}$
b) $B = \{1^{n^2} | n \geq 0\}$
c) $C = \{1^{n^3} | n \geq 0\}$
d) $D = \{1^{n^4} | n \geq 0\}$
e) None